GUNRAAJ SINGH

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PROFESSIONAL PROFILE

Analytical **Electrical Engineering** student with 2+ years of academic and leadership experience in **EV Powertrain and Power Electronics**. Proven ability to design efficient DC-DC converters and conduct complex grid-stability simulations using **MATLAB/Simulink**. Committed to advancing sustainable energy infrastructure through model-based design and hardware-in-the-loop optimization.

EDUCATION

B.Tech in Electrical Engineering — Expected May 2027
Pandit Deendayal Energy University (PDEU) — *Gandhinagar, India*
CGPA: 8.26 / 10.0
Relevant Coursework: Power Electronics, Power System Protection, Electric Drives, Control Systems, Linear Control Theory, Electric Vehicle Technology.

TECHNICAL SKILLS

- **Engineering Domains:** Power Electronics, EV Powertrain, Grid Protection, Control Systems, Smart Grids.
- **Simulation & Design:** MATLAB/Simulink (Simscape), PSIM, PowerWorld, MiPower, PSCAD, Proteus.
- **Programming & Industrial:** C/C++, Python, Embedded C (8051/ARM), SQL, PLC (Ladder Logic).
- **Hardware & Lab:** IR2110 Gate Driving, MOSFET Switching, Oscilloscopes, Logic Analyzers, Sensor Interfacing.

KEY PROJECTS

Vibration Analysis & Diagnostic System (Python, OpenCV)

- Engineered a computer-vision-based vibration analyzer using **Lucas-Kanade optical flow** to track 100+ structural features in real-time.
- Developed a signal processing pipeline involving **FFT** and **Butterworth filters** to extract natural frequencies for mechanical anomaly detection.
- Accomplished sub-pixel tracking accuracy (measured by 0.99 variance thresholding) resulting in a 40% reduction in fault detection latency.

Synchronous Buck Converter Design (Power Electronics Hardware)

- Designed and fabricated a synchronous buck converter utilizing **IR2110 gate drivers** and **IRFZ44N MOSFETs**, achieving 92% efficiency in hardware benchmarking.
- Optimized PCB layout to minimize parasitic inductance, improving transient response stability by 15%.

PMBLDC Motor Speed Control (EV Systems Simulation)

- Developed a closed-loop speed control model in **MATLAB/Simulink** incorporating thermal-aware feedback for real-world EV performance study.
- Maintained speed regulation within ±2% under dynamic load steps across 50+ simulated drive cycle iterations.

EXPERIENCE

Vice President — June 2025 – Present
Anirveda — The TechnoEconomics Club, PDEU — *Gandhinagar, India*

- Spearheaded technical operations for a 30+ member team, orchestrating 10+ annual events for 500+ participants.
- Secured sponsorships by delivering technical value propositions to industry stakeholders.
- Mentored 15+ juniors in **MATLAB and C++** for engineering applications, improving project completion rates by 35%.